

Esolution

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Statistics for Business Administration

Exam: MA9712 / Final-MA9712-22

Date: Thursday 11th August, 2022

Examiner: Prof. Mathias Drton

Time: 17:15 – 18:45

Working instructions

- This exam consists of **18 pages** with a total of **9 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 52 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **scientific pocket calculator**
 - the **cheat sheet**, which is contained in this exam on page **16**
- Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____ / Early submission at _____

d) Consider the joint probability distribution defined through the above contingency table. Does there appear to be a relationship between opinion and age? Clearly state any probabilities you use as justification.

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2

To answer this question we must calculate the probabilities of various opinions based on age:

$$P(\text{Should} | 18-34 \text{ year old}) = \frac{64}{146} \approx 0.44$$

$$P(\text{Should} | 35-49 \text{ year old}) = \frac{55}{118} \approx 0.47$$

$$P(\text{Should} | 50-64 \text{ year old}) = \frac{88}{138} \approx 0.64$$

$$P(\text{Should} | 65+ \text{ year old}) = \frac{65}{98} \approx 0.66$$

Since these probabilities are different, there does appear to be a relationship between opinion on whether Apple should comply with the court order or not and age.

e) Consider the joint probability distribution defined through the above contingency table. Show, that the probability of a randomly selected 18-34 year old thinks Apple should not comply with the court order is equal to 0.342 (rounded to three decimal places).

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We consider the distribution defined through the first column. Hence, we get

$$P(\text{Should not} | 18-34 \text{ year old}) = \frac{50}{146} \approx 0.342$$

f) What is the probability that in a random sample of five 18-34 year olds exactly two believe that Apple should not comply with the court order?

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1

Hint:

```
dbinom(2, size = 5, prob = c(0.292, 0.342, 0.438))
## [1] 0.303 0.333 0.341
```

```
pbinom(2, size = 5, prob = c(0.292, 0.342, 0.438))
## [1] 0.847 0.777 0.615
```

The probability we are looking for is

$$P(K = 2) = \binom{5}{2} \cdot 0.342^2 \cdot (1 - 0.342)^3$$

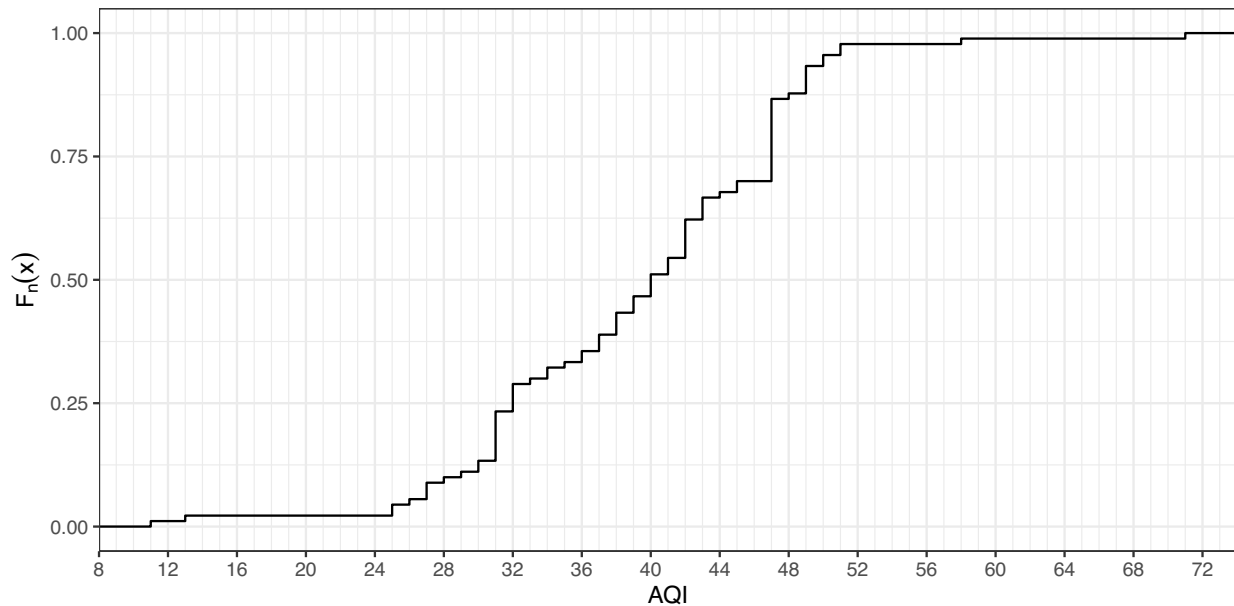
which is due to the hint equal to

```
dbinom(2, size = 5, prob = 0.342)
## [1] 0.333
```

Problem 2 (6 credits)

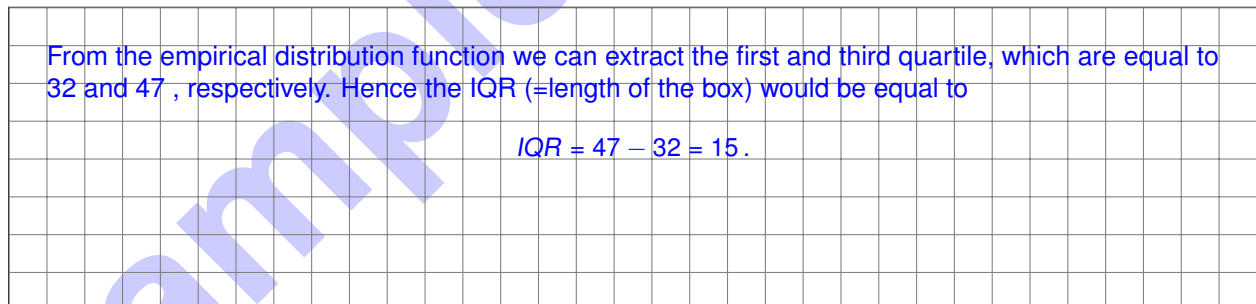
Daily air quality is measured by the air quality index (AQI) reported by the Environmental Protection Agency. This index tells you how clean or polluted your air is, and what associated health effects might be a concern for you. The index is calculated for five major air pollutants regulated by the Clean Air Act, one of which is particulate matter. Values of this index range from 0 to 300 and a higher value indicates lower air quality.

AQI was reported for 90 randomly sampled days in Durham over the past 5 years. The empirical distribution function F_n below shows the distribution of these values. These 90 days represent a reasonably random and representative sample, and you may assume that the AQI levels for these days are independent.



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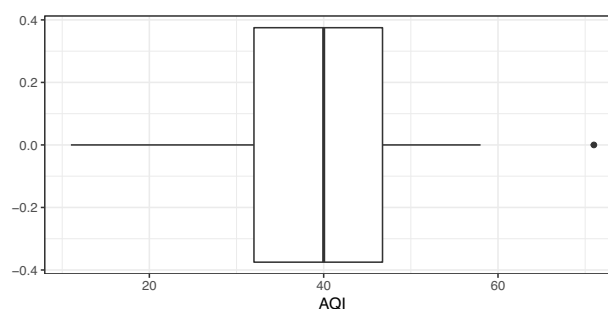
a) Compute the interquartile range.



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b) We would like to use these data to construct a confidence interval for the average AQI in Durham. Evaluate, in context of the data, whether the conditions for inference based on the central limit theorem are reasonably met.

Hint:



The conditions are

- Independence: it is said that we have a random sample with independent observations
- Sample size / skew: From the boxplot we can see, that the distribution of the sample is fairly symmetric. In addition we have $n = 90 \geq 30$, hence we can assume that the population distribution should be symmetric as well. Therefore, we would expect the sampling distribution of average AQI to be nearly normal.

c) The empirical mean and standard deviation of these 90 AQI values is 39.26 and 9.13, respectively. Justify that a two-sided 90% confidence interval for the true average AQI in Durham is equal to (37.68, 40.84) (rounded to two decimal places).

Hint:

```
qnorm(c(0.025, 0.05, 0.1, 0.9, 0.95, 0.975))
## [1] -1.96 -1.64 -1.28 1.28 1.64 1.96
```

The confidence interval is computed with the formula $\bar{x} \pm \text{critical value} \cdot SE$. Using the given estimates and the critical value

```
qnorm(0.95)
## [1] 1.64
```

we obtain the interval

$$39.26 \pm 1.64 \cdot \frac{9.13}{\sqrt{90}} = (37.681686, 40.838314) \\ \approx (37.68, 40.84),$$

which coincides with the given interval.

d) Interpret the confidence interval from the previous part in the context of the data.

We are 90% confident that the average AQI in Durham is between 37.67 and 40.85.

Problem 3 (4 credits)

Jose uses the parking lot at the train station Olympiazentrum every Thursday evening. However, some days the parking lot is full, often due to events in the nearby Olympiapark. There are cultural events on 33% of evenings, sporting events on 22% of evenings, and no events on 45% of evenings. When there is a cultural event, the parking lot fills up about 21% of the time, and it fills up 75% of evenings with sporting events. On evenings when there are no events, it only fills up about 4% of the time.

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4 ☐

If Jose comes to the parking lot and finds it full, what is the probability that there is a sporting event? Write down all steps, which are needed to compute the wanted probability. In case you use a tree, properly label the nodes in the tree.

The outcome of interest is whether there is a sporting event (we call this event A_1), and the condition is that the parking lot is full, which we denote by B . Let A_2 represent an cultural event and A_3 represent there being no event. Then the given probabilities can be written as

$$P(A_1) = 0.22 \quad P(A_2) = 0.33 \quad P(A_3) = 0.45$$

$$P(B|A_1) = 0.75 \quad P(B|A_2) = 0.21 \quad P(B|A_3) = 0.04$$

Bayes' Theorem can be used to compute the probability of a sporting event (A_1) under the condition that the parking lot is full (B):

$$P(A_1|B) = \frac{P(B|A_1) \cdot P(A_1)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3)}$$

$$= \frac{0.75 \cdot 0.22}{0.75 \cdot 0.22 + 0.21 \cdot 0.33 + 0.04 \cdot 0.45} = 0.653983$$

Problem 4 (5 credits)

In a February 2016 Public Policy Polling survey, respondents were asked which team they would root for at the next basketball game between Duke University and University of North Carolina (UNC). The responses are summarized below.

```
addmargins(table(duke_unc))
```

##	Team			
##	Sex	Duke	UNC	Sum
##	Man	188	225	413
##	Woman	178	287	465
##	Sum	366	512	878

0 ☐

1 ☐

a) We would like to test if there is a difference between the proportions of men and women who will root for Duke on the next basketball game between these two teams using a randomization test. State the null and alternative hypothesis of this test problem.

Let $p_i, i \in \{M, W\}$, be the proportion of men/women who root for Duke. Then the test problem is given by

$$H_0 : p_M = p_W \quad H_A : p_M \neq p_W.$$

b) Show that the observed sample statistic is equal to 0.072 (rounded to three decimal places).

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The observed proportions are equal to

$$\hat{p}_M = \frac{188}{413} \quad \text{and} \quad \hat{p}_W = \frac{178}{465},$$

and the difference is given by

$$\hat{p}_M - \hat{p}_W = \frac{188}{413} - \frac{178}{465} = 0.07241011 \approx 0.072,$$

which is equal to the given value.

c) The simulated null distribution was computed using the following code

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1

```
library(infer)
null_dist <- duke_unc %>%
  specify(Team ~ Sex, success = "Duke") %>%
  hypothesize(null = "independence") %>%
  generate(reps = 200) %>%
  calculate(stat = "diff in props", success = "Duke", order = c("Man", "Woman"))
```

Calculate the p-value using the following output

```
null_dist %>%
  filter(abs(stat) >= 0.072) %>%
  group_by(stat) %>%
  summarize(n = n())
```

```
## # A tibble: 6 x 2
##   stat      n
##   <dbl> <int>
## 1 -0.0922     1
## 2 -0.0830     1
## 3 -0.0785     1
## 4  0.0770     1
## 5  0.0816     1
## 6  0.0861     1
```

From the output we detect, that 6 of the 200 simulated differences were as large or larger than the observed differences. Hence we get a p-value of $\frac{6}{200} = 0.03$.

d) Using $\alpha = 0.05$, state the conclusion of the hypothesis test. Justify your answer. Interpret the conclusion in context of the data and the hypotheses.

0
1

Since the p-value is equal to 0.03 and thus less than the significance level 0.05, we reject H_0 . The data provide convincing evidence of a difference between the proportions of men and women who will root for Duke on the next basketball game between these two teams.

Problem 5 (4 credits)

Determine if the following statements are **true** or **false**. Justify your answer

Remark: Only correct justifications are awarded with points. Not the decision (true/false).

- 0 ☐
1 ☐
- a) In order for the sampling distribution of a mean to be nearly normally distributed, the expected numbers of successes and failures must be at least 10.

False: The success-failure condition is only relevant for the mean of binary variable.

- 0 ☐
1 ☐
- b) Random sampling is sufficient for finding evidence of a causal relationship.

False: Besides random sampling we additionally need random assignment of the treatment for having the chance of finding causal relationships.

- 0 ☐
1 ☐
- c) Suppose a researcher sets the significance level at 5% prior to conducting an experiment. After conducting the study, the researcher computes a p-value of 0.04. The researcher can be certain that the null hypothesis is false.

False: The data provided convincing evidence for rejecting the null hypothesis. But the researcher could still be committing a type I error. Hence, he can't conclude that the null hypothesis is for sure false.

- 0 ☐
1 ☐
- d) If a given value (for example, the null hypothesized value of a parameter) is within a 90% confidence interval for a population parameter, it will definitely also be within a 95% confidence interval for the same population parameter calculated based on the same sample.

True: With increasing confidence level the confidence interval has to become wider, because otherwise it would not be able to cover the true population parameter with the specified probability.

Problem 6 (3 credits)

A psychologist wants to determine if socio-economic status is related to game playing preferences. Fifty children, in total, were identified from families of low, middle, and high socio-economic status, and then the children were asked to select one of the games: Monopoly, Battleship, or Connect Four. They performed a chi-squared test which yielded a test statistic value of 5.0.

a) State the null hypothesis H_0 of the chi-squared test.

0
1

The null hypothesis is equal to

H_0 : socio-economic status and game playing preferences are independent.

b) State a range (p^L, p^U) for the p-value of the chi-squared test, which is as small as possible, using the values in the hint.

Hint:

```
qchisq(c(0.05, 0.1, 0.2, 0.3), df = 3, lower.tail = FALSE)
## [1] 7.81473 6.25139 4.64163 3.66487
```

```
qchisq(c(0.05, 0.1, 0.2, 0.3), df = 4, lower.tail = FALSE)
## [1] 9.48773 7.77944 5.98862 4.87843
```

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Since both variables have three levels, the chi-square test uses a χ^2 -distribution with 4 degrees of freedom. Therefore, we can define the following range

$0.2 = P_{H_0}(\chi^2 > 5.988617) < P_{H_0}(\chi^2 > 5.0) < P_{H_0}(\chi^2 > 4.878433) = 0.3,$

i.e.

$p\text{-value} \in (0.2, 0.3).$

Problem 7 (9 credits)

In a study that investigates the impact of the Danish Action Plan for the Aquatic Environment, which addresses pollution of the Danish water resources, the concentration of nitrogen (measured in g/m^3) was measured in a particular river in 1998, 2003, and 2011. Six measurements were randomly taken in each year. A summary of the nitrogen concentration is provided.

```
library(tidyverse)
AE %>%
  group_by(year) %>%
  summarize(mean = mean(nitrogen),
            sd = sd(nitrogen),
            n = n()
  )
```

```
## # A tibble: 3 x 4
##   year   mean    sd     n
##   <chr> <dbl> <dbl> <int>
## 1 1998   5.55 0.486     6
## 2 2003   5.19 0.362     6
## 3 2011   4.05 0.673     6
```

We want to test at the 5% significance level the relationship between the two variables using an ANOVA model.

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1 ☐

a) State the null and alternative hypotheses for testing the relationship between the two variables?

H_0 : Average nitrogen concentration does not vary across years
 H_A : Average nitrogen concentration does vary across years, there is at least one mean different from the rest (at least two means are different)

0 ☐
1 ☐
2 ☐

b) What are the conditions for ANOVA? Evaluate whether each of these conditions are satisfied in the context of these data. **Note any additional information you would need** to fully evaluate the conditions.

Independence: The samples were randomly taken. But the description does not give any information about how the sampling locations were chosen. Nevertheless we can assume each measurement in this sample is independent of another with respect to nitrogen concentration
Normality: With the given information, this is hard/not to say. We would need more info (like a plot) to confirm this assumption.
Constant variance: The standard deviations aren't very similar, sample size is also small, so this could be a problem.

c) Replace the missing values (NA) in the following ANOVA table. Write the missing values in the table given in the box. Only the values added in the table will be graded.

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```
summary(aov(nitrogen~year, data=AE))
      Df Sum Sq Mean Sq F value Pr(>F)
year    NA    7.36      NA    13.47 0.00045 ***
Residuals NA      NA      NA
---
```

Hint:

```
var(AE$nitrogen) * 17
## [1] 11.46
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
summary(aov(nitrogen~year, data=AE))
      Df      Sum Sq   Mean Sq    F value    Pr(>F)
year      2      7.36      3.68      13.47    0.00045 ***
Residuals 13     4.10      0.31
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

DF = (2, 15) , SumSq =(7.36, 11.46 -7.36=4.1), MeanSq = (7.36/2, 4.1/15)

d) State the conclusion of the hypothesis test. Justify your answer. Interpret the conclusion in context of the data and the hypotheses.

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Reject the null hypothesis at the 5% level since the p-value is less than 0.05. The data provide evidence that the average nitrogen concentration is different for at least one pair of years, but may not be convincing because of the SD condition.

e) What percent of total variability in nitrogen concentration is explained by year?

0
1

The total variability in nitrogen concentration explained by year is equal to

$$\frac{SS_G}{SS_T} = \frac{7.36}{11.46} \approx 0.64.$$

f) Assume you want to conduct pairwise t-tests in a post-hoc analysis using a Bonferroni correction. Determine the significance level, which should be used for each of these t-tests.

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1/2

Since we need to conduct $\binom{3}{2} = 3$ t-tests, the significance level for each of these tests should be $\alpha^* = \frac{0.05}{3} = 0.0167$.

Problem 8 (8.5 credits)

The Child Health and Development Studies considered in a study all pregnancies between 1960 and 1967 among women in the Kaiser Foundation Health Plan in the San Francisco East Bay area. Therefore, they collected a random sample of 800 mothers. Variables of interest include bwt (birth weight of the child in ounces), parity (1 - first born child, 0 - otherwise), gestation (length of pregnancy in days), age (mother's age in years), height (mother's height in inches), weight (mother's pregnancy weight in pounds) and smoke (1 - mother is a smoker, 0 - otherwise).

Call:

```
lm(formula = bwt ~ ., data = babies)
```

Residuals:

Min	1Q	Median	3Q	Max
-54.249	-10.315	-0.135	10.269	51.463

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-108.46339	17.29916	-6.270	5.93e-10	***
gestation	0.53578	0.03701	14.476	< 2e-16	***
parity	-3.65823	1.33525	-2.740	0.00629	**
age	0.11144	0.10270	1.085	0.27819	
height	1.15701	0.24055	4.810	1.81e-06	***
weight	0.04087	0.03129	1.306	0.19196	
smoke	-8.07276	1.14557	-7.047	3.98e-12	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.72 on 793 degrees of freedom

Multiple R-squared: NA, Adjusted R-squared: NA

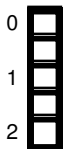
F-statistic: 54.48 on 6 and 793 DF, p-value: < 2.2e-16



a) State the equation of the estimated model.

Remark: You can round the estimated coefficients to two decimal places.

$\widehat{bwt} = -108.46 + (0.54 \cdot \text{gestation}) + (-3.66 \cdot \text{parity}) + (0.11 \cdot \text{age}) + (1.16 \cdot \text{height}) + (0.04 \cdot \text{weight}) + (-8.07 \cdot \text{smoke})$																			
--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



b) Interpret the estimated slopes for gestation and smoke.

Interpretation of the slope for gestation: All else equal, the birth weight will increase on average by 0.54 ounces for each additional day of the pregnancy.																			
Interpretation of the slope for smoke: All else equal, the birth weight will on average be 8.07 ounces lower in case the mother smokes compared to non-smoking mothers.																			

c) State the null and alternative hypothesis, which are tested on the line where the p-value is 0.006287.

Null and alternative hypothesis for the partial t-test with p-value 0.006287 are

$$H_0 : \beta_{\text{parity}} = 0 \quad H_A : \beta_{\text{parity}} \neq 0.$$

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d) Calculate the adjusted R^2 value.

Hint:

```
var(babies$bwt)
## [1] 346.472
```

The formula for the adjusted R^2 value is

$$R_{adj}^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2 / (n - k - 1)}{\sum_{i=1}^n (y_i - \bar{y})^2 / (n - 1)}.$$

From the summary we get $\sum_{i=1}^n (y_i - \hat{y}_i)^2 / (n - k - 1) = 15.72^2$ and the hint says $\sum_{i=1}^n (y_i - \bar{y})^2 / (n - 1) = 346.472$. Hence, the adjusted R^2 is equal to

$$R_{adj}^2 = 1 - \frac{15.72^2}{346.472} = 0.286758.$$

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e) Now we apply the AIC approach to reduce the model with a backward elimination. Use the following output to decide if a variable can be dropped in the first step of the backward elimination. Justify your answer.

Remark: Be aware that the given output is not the standard output of `step()`. The rows have been permuted.

```
step(model)
Start: AIC=4415.11
bwt ~ gestation + parity + age + height + weight + smoke
```

	Df	Sum of Sq	RSS	AIC
- gestation	1	51803	247828	4600.7
- height	1	5719	201744	4436.1
- weight	1	422	196447	4414.8
<none>			196026	4415.1
- parity	1	1855	197881	4420.6
- age	1	291	196317	4414.3
- height	1	5719	201744	4436.1
- smoke	1	12276	208301	4461.7

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We can drop a variable, if this leads to a decrease in AIC. This is the case for age and weight. Since age leads to the larger decrease, we would drop age in the first step.

Problem 9 (5 credits)

The RMS Titanic sank in the early morning hours of 15 April 1912 in the North Atlantic Ocean. The dataset titanic contains information about passengers aboard of the RMS Titanic

titanic

```
## # A tibble: 714 x 7
##   Survived Pclass Sex      Age SibSp Parch  Fare
##   <int>   <int> <chr>   <dbl> <int> <int> <dbl>
## 1       0     3 male     22     1     0  7.25
## 2       1     1 female   38     1     0 71.3
## 3       1     3 female   26     0     0  7.92
## 4       1     1 female   35     1     0 53.1
## 5       0     3 male     35     0     0  8.05
## 6       0     1 male     54     0     0 51.9
## 7       0     3 male      2     3     1 21.1
## 8       1     3 female   27     0     2 11.1
## 9       1     2 female   14     1     0 30.1
## 10      1     3 female    4     1     1 16.7
## # ... with 704 more rows
```

We are interested in predicting the survival of the passenger (Survived - 1="yes", 0="no") using the variables Pclass (Passenger Class), Sex, Age, SibSp (number of sibling/spouses aboard), Parch (number of parents/children aboard) and Fare as predictor variables.

We split the data into a training set titantic_train and a test set titantic_test.

- 0 ☐ a) Since we have binary response variable, a logistic regression model will be fitted to the data. Add the missing part to the following code, such that a logistic regression model using all variables as predictor variables will be fitted to the training data.

```
model_full <- glm(Survived ~ ., data = titantic_train, ___)
```

```
model_full <- glm(Survived ~ ., data = titantic_train,
                  family = binomial )
```

- 0 ☐ b) A summary of the fitted model is the following

```
## # A tibble: 7 x 5
##   term      estimate std.error statistic  p.value
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)  4.77      0.674      7.08 1.41e-12
## 2 Pclass     -0.997    0.178     -5.59 2.28e- 8
## 3 Sexmale    -2.71     0.244    -11.1 1.40e-28
## 4 Age        -0.0415   0.00932   -4.46 8.24e- 6
## 5 SibSp      -0.407    0.142     -2.87 4.05e- 3
## 6 Parch      -0.0736   0.138     -0.534 5.93e- 1
## 7 Fare        0.00270 0.00262    1.03 3.02e- 1
```

Give an interpretation of the estimated slope of Sex in terms of the odds ratio.

The odds ratio for a change from female to male is equal to $e^{-2.71} \approx 0.07$, i.e. that the odds of surviving the Titanic disaster is decreased by almost 93%, given all else being equal.

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c) Given the full model, we are now able to compute predictions for test dataset.

```
titanic_test$pred_prob <- predict(model_full, newdata = titanic_test, type = "response")
```

To predict survival we need a threshold for the computed probabilities of survival. Choosing a threshold of 0.5 leads to the following table

```
table(titanic_test$Survived, titanic_test$pred_prob > 0.5)
```

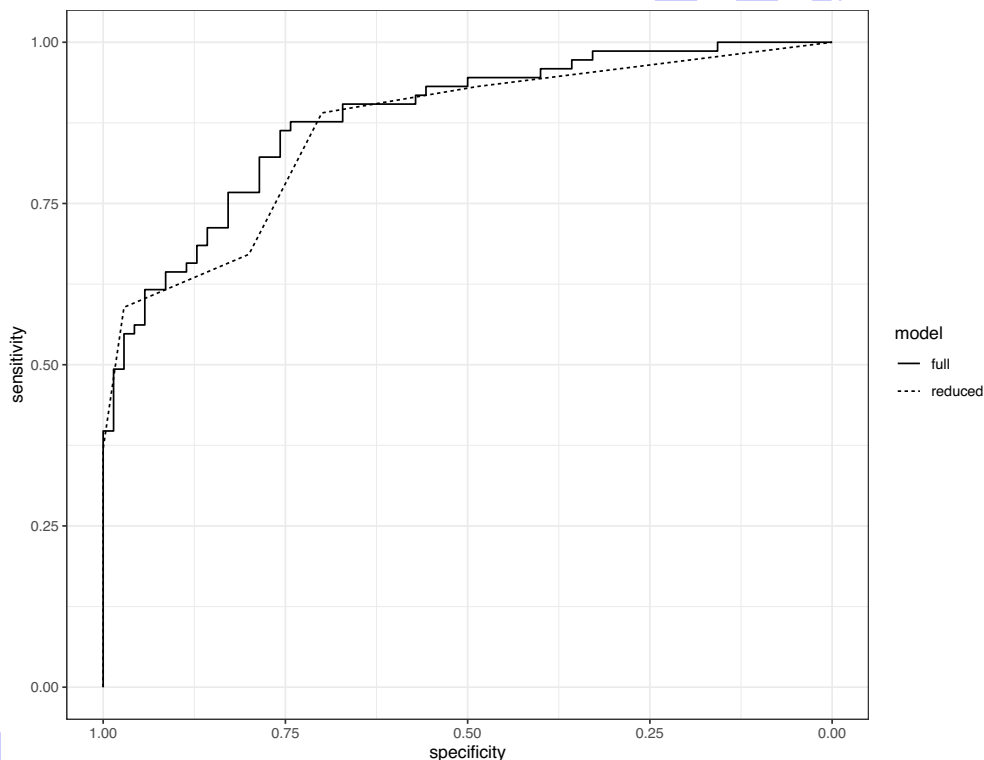
```
##
##      FALSE TRUE
##  0      58   12
##  1      18   55
```

Compute the sensitivity of this prediction.

The sensitivity of this prediction is equal to $\frac{55}{73} = 0.753425$.

d) Finally we compare the predictive accuracy of the model to a reduced model containing just the variables Sex and Pclass. The following graphic contains the ROC curves of the full and reduced model.

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2



The area under the ROC curve of the full model is equal to 0.8818 and the reduced model leads to an area under the ROC curve of 0.8641. Which of the two models shows the better predictive accuracy with respect to the ROC curve? Justify your answer.

The larger the area under the ROC curve, the better the predictive accuracy. Since the full model has a larger auc value, we would pick the full model.

Cheat sheet

Summarizing data:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad s = \sqrt{s^2} \quad IQR = Q_3 - Q_1 \quad Q_1 - 1.5 \cdot IQR, \quad Q_3 + 1.5 \cdot IQR$$

Probability:

$$P(A^c) = 1 - P(A); \quad P(A \cap B) = P(A) \cdot P(B), \quad A, B \text{ independent}; \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)};$$

$$E(X) = \sum_{i=1}^k x_i P(X = x_i), \quad E(X) = \int_{\mathbb{R}} x \cdot f(x) dx; \quad \text{Var}(X) = E(X^2) - E(X)^2;$$

$$SD(X) = \sqrt{\text{Var}(X)}; \quad \text{Cov}(X, Y) = E(X \cdot Y) - E(X) \cdot E(Y); \quad Z = \frac{X - E(X)}{\sqrt{\text{Var}(X)}}$$

$$\text{Geom. dist.: } P(X = x) = (1 - p)^{x-1} p, \quad x \in \mathbb{N}; \quad E(X) = \frac{1}{p}, \quad SD(X) = \sqrt{\frac{1-p}{p^2}}$$

$$\text{Binom. dist.: } P(X = x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}, \quad x \in \{0, 1, 2, \dots, n\}; \quad E(X) = np, \quad SD(X) = \sqrt{np(1-p)}$$

$$\text{Neg. Binom. dist.: } P(X = x) = \frac{(x-1)!}{(k-1)!(x-k)!} p^k (1-p)^{x-k}, \quad x \in \mathbb{N};$$

$$\text{Pois. dist.: } P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x \in \mathbb{N}_0; \quad E(X) = \lambda, \quad SD(X) = \sqrt{\lambda}$$

Inference for categorical data:

$$\hat{p} \pm z^* \cdot SE_{\hat{p}} \quad SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \quad Z = \frac{\hat{p} - p_0}{SE_{\hat{p}}} \quad SE_0 = \sqrt{\frac{p_0(1-p_0)}{n}}$$

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \cdot SE \quad SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \quad Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{SE_0} \quad SE_0 = \sqrt{\hat{p}_{pooled}(1-\hat{p}_{pooled}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$\hat{p}_{pooled} = \frac{\sum x_{1,i} + \sum x_{2,i}}{n_1 + n_2}; \quad \chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}, \quad \text{expected} = \frac{\text{row} \times \text{column}}{\text{total}} \text{ (ind.)}, \quad df = k - 1 \text{ or } (r-1) \cdot (c-1)$$

Inference for numerical data:

$$\bar{x} \pm t_{df}^* \cdot SE_{\bar{x}} \quad T_{df} = \frac{\bar{x} - \mu_0}{SE_{\bar{x}}} \quad SE_{\bar{x}} = \frac{s}{\sqrt{n}} \quad df = n - 1$$

$$(\bar{x}_1 - \bar{x}_2) \pm t_{df}^* \cdot SE \quad T_{df} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE} \quad SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \quad df = \min(n_1 - 1, n_2 - 1)$$

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \quad df = n_1 + n_2 - 2$$

ANOVA and linear regression:

$$F = \frac{MSG}{MSE} \quad df_1 = k - 1, \quad df_2 = n - k, \text{ for } k \text{ groups}; \quad T_{df} = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{\frac{MSE}{n_i} + \frac{MSE}{n_j}}} \quad df = n - k \quad \alpha^* = \alpha / K$$

$$\hat{y} = b_0 + b_1 x \quad b_1 = R_{xy} \cdot \frac{s_y}{s_x} \quad \bar{y} = b_0 + b_1 \bar{x} \quad \text{residual} = y - \hat{y}$$

$$\hat{\sigma} = \sqrt{\frac{1}{n-2} \sum (y_i - \hat{y}_i)^2} = \sqrt{\frac{1}{n-2} \sum (y_i - b_0 - b_1 x_i)^2}$$

$$R_{xy} = \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{s_x \cdot s_y} \quad T_{df} = \frac{b_1 - \beta_{1,0}}{SE_{b_1}} \quad b_1 \pm t_{df}^* \cdot SE_{b_1} \quad df = n - 2$$

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}; \quad R_{adj}^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2 / (n - k - 1)}{\sum_{i=1}^n (y_i - \bar{y})^2 / (n - 1)};$$

$$AIC = \text{constant} + n \cdot \ln \left(\sum_{i=1}^n e_i^2 \right) + 2k$$

Logistic regression:

$$\text{logit}(p_i) = \log \left(\frac{p_i}{1 - p_i} \right) = \beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i} \quad OR_j = \frac{\text{odds when } x_{i,j} = x+1}{\text{odds when } x_{i,j} = x} = e^{\beta_j}$$

$$\text{sensitivity} = P(\text{Test} + | \text{Condition} +), \quad \text{specificity} = P(\text{Test} - | \text{Condition} -)$$

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

Sample Solution

Sample Solution